

Counting a Repeating Pattern of Length 3

Suppose we have a sequence where the same pattern repeats again and again. For example, let the repeating pattern be

$$A, B, C, A, B, C, A, B, C, \dots$$

Now we want to count how many times A , B , and C appear in the first n positions.

The main trick is to divide the sequence into complete blocks of length 3. Each complete block

$$[A, B, C]$$

contains exactly one A , one B , and one C .

Let

$$q = \left\lfloor \frac{n}{3} \right\rfloor$$

and

$$r = n \bmod 3$$

Here, q is the number of complete blocks, and r is the number of extra elements after taking all complete blocks.

From the q complete blocks, each of A , B , and C appears exactly q times. So initially,

$$cnt_A = q, \quad cnt_B = q, \quad cnt_C = q$$

Now we only need to handle the remaining r elements.

If $r = 0$, then there are no extra elements:

$$cnt_A = q, \quad cnt_B = q, \quad cnt_C = q$$

If $r = 1$, then the extra element is A :

$$cnt_A = q + 1, \quad cnt_B = q, \quad cnt_C = q$$

If $r = 2$, then the extra elements are A and B :

$$cnt_A = q + 1, \quad cnt_B = q + 1, \quad cnt_C = q$$

For example, let

$$n = 17$$

Then

$$q = \left\lfloor \frac{17}{3} \right\rfloor = 5$$

and

$$r = 17 \bmod 3 = 2$$

So there are 5 complete blocks. From those blocks,

$$cnt_A = 5, \quad cnt_B = 5, \quad cnt_C = 5$$

Since $r = 2$, the two extra elements are A and B . Therefore,

$$cnt_A = 6, \quad cnt_B = 6, \quad cnt_C = 5$$

So in the first 17 positions,

A appears 6 times

B appears 6 times

C appears 5 times